

LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 2
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1. TITLE

Webpage Styling with Canvas and CSS3

2. AIM

To structure and style a webpage using HTML5 canvas shapes, CSS gradients, card layouts, shadows, hover transitions, and navigation bars.

3. OBJECTIVE

The objectives of this experiment are:

- To implement linear gradient backgrounds and text shadow effects.
- To style interactive navigation bar with hover color changes.
- To construct card containers with rounded borders and drop shadows.
- To program HTML5 Canvas 2D context for drawing rectangles, circles, lines, and text.

4. THEORY

CSS3 provides gradients, text shadows, box shadows, and pseudo-classes (:hover) for modern visual design. HTML5 Canvas element allows dynamic 2D graphic rendering via JavaScript APIs.

Application Flow:

HTML5 Markup -> CSS3 Styling -> Navigation & Cards -> Canvas 2D Context API -> Rendered Webpage

5. ALGORITHM / PROCEDURE

1. Create task2.html with header, navigation bar, canvas element, card containers, and footer.
2. Apply CSS linear-gradient on body and styled header with text shadow.
3. Add navigation links with :hover state changing color to yellow.
4. Style .card class with rounded corners, padding, and box shadow.
5. Access canvas 2D context in JavaScript and render rectangle, arc, line, and fillText.

6. PROGRAM

task2.html

```
<!DOCTYPE html>
<html>

<head>

<title>My Styled Webpage</title>

<style>

body{
    background:linear-gradient(lightblue,white);
    font-family:Arial;
}

header{
    background-color:blue;
    color:white;
    text-align:center;
    padding:15px;
    text-shadow:2px 2px gray;
}

nav{
    background-color:black;
    padding:10px;
    text-align:center;
}

nav a{
    color:white;
    text-decoration:none;
    margin:15px;
}

nav a:hover{
    color:yellow;
}

canvas{
    border:2px solid black;
    background-color:white;
}

.card{
    border:2px solid black;
    border-radius:10px;
    background-color:white;
    margin:20px;
    padding:15px;
    box-shadow:4px 4px 5px gray;
}

h2{
    font-family:Georgia;
}

p{
    font-size:16px;
```

```
}

button{
  background-color:blue;
  color:white;
  padding:8px;
  border:none;
}

button:hover{
  background-color:green;
}

footer{
  background-color:black;
  color:white;
  text-align:center;
  padding:10px;
}

</style>

</head>

<body>

<header>

<h1>My Styled Webpage</h1>

<p>HTML5 Canvas and CSS3 Example</p>

</header>

<nav>

<a href="#">Home</a>

<a href="#">About</a>

<a href="#">Services</a>

<a href="#">Contact</a>

</nav>

<br>

<center>

<h2>Canvas Example</h2>

<canvas id="myCanvas" width="400" height="220"></canvas>

</center>

<div class="card">

<h2>HTML</h2>

<p>HTML is used to create web pages.</p>
```

```
<button>Read More</button>

</div>

<div class="card">

<h2>CSS</h2>

<p>CSS is used to style web pages.</p>

<button>Learn</button>

</div>

<div class="card">

<h2>Canvas</h2>

<p>Canvas is used to draw shapes and text.</p>

<button>Example</button>

</div>

<footer>

<p>Designed by Dinesh Kumar</p>

</footer>

<script>

var c=document.getElementById("myCanvas");
var ctx=c.getContext("2d");

ctx.fillStyle="red";
ctx.fillRect(30,30,100,60);

ctx.beginPath();
ctx.arc(220,60,30,0,2*Math.PI);
ctx.fillStyle="blue";
ctx.fill();

ctx.beginPath();
ctx.moveTo(30,150);
ctx.lineTo(350,150);
ctx.stroke();

ctx.font="20px Arial";
ctx.fillStyle="black";
ctx.fillText("Dinesh Kumar",120,190);

</script>

</body>

</html>
```

7. CODE EXPLANATION

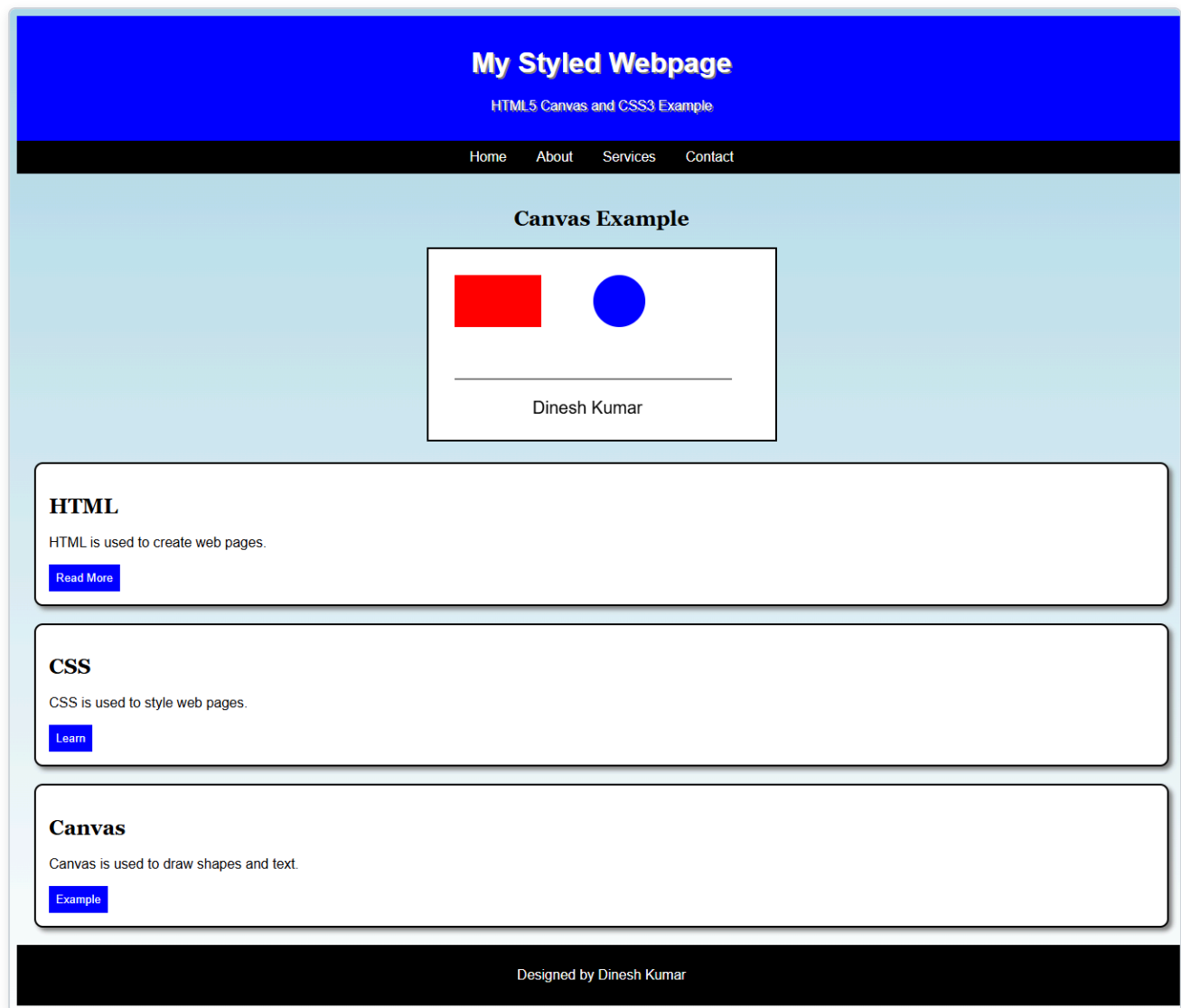
Code / Syntax	Explanation
<code>linear-gradient(lightblue, white)</code>	Creates smooth background color gradient transition.
<code>text-shadow: 2px 2px gray</code>	Adds subtle drop shadow behind header text.
<code>nav a:hover { color: yellow; }</code>	Changes navigation link color to yellow on mouse hover.
<code>box-shadow: 4px 4px 5px gray</code>	Applies realistic box shadow depth to card elements.
<code>ctx.fillRect(30, 30, 100, 60)</code>	Draws a solid filled rectangle on 2D canvas context.
<code>ctx.arc(220, 60, 30, 0, 2*Math.PI)</code>	Draws a circular arc path on canvas.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Load	Open task2.html	Header gradient & navigation render properly	Pass
TC-02	Nav Link Hover	Hover over links	Link text changes color to yellow	Pass
TC-03	Canvas Drawing	Inspect canvas	Rectangle, circle, line & text render correctly	Pass
TC-04	Card Layout	Inspect card containers	Three cards render with borders & box shadows	Pass

9. OUTPUT



Output 1: Webpage Styling with Canvas and CSS3 Rendered Webpage

10. RESULT

Thus, webpage styling with canvas and css3 was successfully implemented and verified across all test cases.